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Original research article

Level of adherence, awareness and challenges to medical waste management program in two hospitals, a comparative study

Maha Hussien Ali Awad Greda, Nayera Samy Mostafa, Doaa Mahmoud Elhossieny, Mostafa Yosef, Mohamed Salah Gabal

Department of Community, Environmental and Occupational Medicine, Faculty of Medicine – Ain Shams University; Abbassia, Cairo, Egypt

ABSTRACT

Background: Healthcare acquired infections are associated with increased morbidity and mortality among hospitalized patients and predisposes healthcare workers (HCWs) to an increased risk of infections. This study explores the level of awareness and adherence of HCWs towards medical waste management (MWM) in two hospitals.

Objective: to compare the level of awareness and adherence of HCWs towards MWM in two hospitals.

Methods: A comparative cross-sectional design was used, involving 200 HCWs from two hospitals: one, a university-affiliated specialized facility located in Al-Obour City; the other, a public hospital under the Ministry of Health located in Al-Abbasia. Data collection tools included a structured self-administrated questionnaire and Likert-scale adherence assessments.

Results: The results revealed that HCWs at the public hospital demonstrated significantly higher awareness and adherence scores compared to those at the university-affilitated specialized one. Cleaning workers, while showing improved scores regarding the level of awareness at the public hospital, reported the lowest adherence levels overall. Awareness and adherence scores were significantly correlated, highlighting the interplay between understanding and execution of proper waste management. Demographic factors, such as age and sex, showed no significant association with awareness and adherence levels.

Conclusion: The study revealed significant differences in MWM practices between the two hospitals, with the public hospital showing better awareness and adherence. The findings highlight the impact of institutional practices

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Correspondence: Dr. Maha Hussien Ali Awad Greda / Department of Community, Environmental and Occupational Medicine, Faculty of Medicine, Ain Shams University, Abbassia, Cairo 11566, Egypt /
E-mail: Mahagareda@med.asu.edu.eg
ORCID: 0009-0004-7919-1694

and continuous training on compliance. These insights can be integrated into the results section to enhance the analysis of MWM practices.

Key words: Medical waste management, adherence, awareness, healthcare workers, waste disposal

Introduction

Medical waste (MW) is defined as waste generated during the diagnosis, testing, treatment, research or production of biological products for humans or animals. Medical waste includes syringes, live vaccines, laboratory samples, body parts, bodily fluids and waste, sharp needles, cultures and lancets (1). MWM includes all activities involved in waste generation, segregation, transportation, storage, treatment and final disposal of all types of waste generated in the healthcare facilities, stages of which require special attention (2). WHO states that of the total amount of waste generated by health-care activities, 85% are generally nonhazardous waste and the remaining 15% are considered hazardous material that may be infectious, toxic, or radioactive (3). High-income countries generate on average up to 0.5 kg of hazardous waste per bed per day, while low-income countries generate on average 0.2 kg. However, MW is often not separated into hazardous or nonhazardous wastes in low-income countries making the real quantity of hazardous waste much higher (4). Poor management of MW exposes HCWs, waste handlers and the community to infections, toxic effects and injuries. There is also a potential for spreading drug-resistant microorganisms from health-care facilities into the environment through poor MWM (3). Hospitals in Egypt produce an average of 77.447 kg/day of MW (6), less than 50% of these amounts are safely and efficiently discarded through incineration (6). The remaining amount is either illegally traded for recycling or mixed with municipal solid waste (5). Lack of awareness about the health hazards related to waste, inadequate training in proper waste management, absence of waste management and disposal systems, insufficient financial and human resources and the low priority given to the topic are the most common problems connected with MW. Many countries either do not have appropriate regulations, or do not enforce them (7). While several studies in Egypt and similar contexts have addressed medical waste management awareness and practices among HCWs, few have examined how institutional type and exposure to high-infection-risk environments influence adherence (3). This study addresses that gap by directly comparing a public fever hospital, which routinely handles infectious cases, with a university-affiliated specialized hospital, where general clinical care dominates. By integrating a comparative element, our research provides novel evidence on how institutional settings shape medical waste management behaviors, beyond individual demographics. This insight is particularly relevant for policy adaptations targeting specialized hospitals, which may lag in practical waste handling despite high academic resources. Unlike prior research which focused on single-hospital assessments or general staff populations, this study stratifies findings by role (doctor, nurse, cleaner) across distinct institutions.

Participants and methods

A comparative cross sectional study was conducted among two hospitals which are a university-affiliated specialized facility located in Al-Obour City and the other, a public hospital under the Ministry of Health located in Al-Abbasia.The study duration was one year, during which data collection, analysis, and reporting were completed.

Sample Size: using EPI info 7 program for sample size calculation, setting confidence level at 95% and two-sided margin of error at 15%. It was estimated that sample size of 175 health care workers would be needed to detect an expected 50% level of awareness regarding MWM. The number of participants was increased to 200 to account for any potential issues or losses during data collection.

Sampling strategy:

A convenience non-random sampling method was used, which may introduce selection bias due to the voluntary nature of participation and potential overrepresentation of more motivated or aware staff. Although this approach facilitated access to a diverse set of healthcare workers across both hospitals, it limits the generalizability of findings. This limitation is further addressed in the discussion section.

Study population

200 HCWS (doctors, nurses and clean workers) from both hospitals were taken by convenience non random sampling.

Equal numbers of doctors, nurses, and clean workers were included in each hospital (40:40:20) to allow for balanced statistical comparisons across job roles. While this distribution may not mirror actual staffing proportions, it was chosen to ensure sufficient representation of each group for subgroup analysis. This design choice is acknowledged as a limitation.

Study setting:

Participants were selected from a range of core clinical departments that were present in both hospitals, ensuring comparability in the type of healthcare services provided and the volume and nature of medical waste generated. These departments included internal medicine, general surgery, pediatrics, obstetrics and gynecology, and emergency units, which are known to consistently generate various types of MW such as sharps, infectious waste, and pathological materials.

By ensuring that participants worked in comparable departments across both facilities, the study controlled for differences in waste exposure and handling demands related to departmental functions. This design strengthened the validity of comparing awareness and adherence levels between the two institutions, focusing the analysis on institutional factors rather than departmental discrepancies.

Study tools

Self-administrated questionnaire

To assess the level of awareness and adherence of HCWs towards medical waste management.

It was designed by the researcher after reviewing the literature (8, 9, 10) and the Egyptian national guidelines for infection control 2020 (11). The questionnaire consisted of three sections. The first section included the socio-demographic characteristics of HCWs. The second section assessed the level of awareness of HCWs towards MWM. The third section assessed the adherence of HCWs towards medical waste management. The awareness items scored either 1 or 0 for the correct or incorrect response, respectively. the total awareness score for each respondent can range from 0 to 16 (8).

The answers for the level of adherence section was in the form of a 3-point Likert scale (always, sometimes and never) and scored as 2,1 and 0 respectively. The total level of adherence score for each respondent can range from 0 to 24 (8).

Questionnaire validation

The questionnaire was developed based on previously published tools (8–10) and the Egyptian national infection control guidelines (2020). To ensure content validity, it was reviewed by three infection control experts. A pilot study involving 20 HCWs (excluded from the main study) was conducted to test clarity and usability. Internal consistency reliability was assessed, and the Cronbach's alpha coefficients were found to be 0.82 for the awareness section and 0.79 for the adherence section, indicating acceptable reliability.

Ethical considerations: Ethical approval was obtained from faculty of medicine, Ain Shams University, Research Ethics committee(under the number code FMASU MD 155 /2022 FWA 000017585).Informed consent was taken from each participant. The confidentiality and privacy of data were assured.

Statistical analysis: Analysis was performed using SPSS for windows version 20. Data was presented in terms of range, mean and standard deviation (for numeric parametric variables); range, median and interquartile range (for numeric non-parametric variables);

or number and percentage (for categorical variables). Difference between two independent groups was analyzed using independent student's t-test as well as the mean difference and its 95% CI (for numeric parametric variables); or chi-squared test as well as the risk ratio and its 95% CI (for categorical variables). (ANOVA) was used to compare more than two groups. P value <0.05was considered statistically significant.

Results

Across all staff jobs (doctors, nurses, and clean workers), comparisons between the two hospitals revealed no significant differences in demographic variables such as age, gender, or work experience (Table 1).

The study revealed that doctors at the public hospital demonstrated significantly higher awareness scores compared to those at the university-affiliated specialized facility (81.9% vs. 76.4%, p = 0.035), Similarly, practice scores among doctors were significantly higher at the public hospital (87.7% vs. 75.2%, p < 0.001), suggesting that enhanced awareness may contribute to improved adherence (Table 2). Among nurses, awareness scores were significantly higher at the public hospital compared to the specialized one (90.3% vs. 85.0%, p = 0.021), and adherence scores were also notably higher (95.0% vs. 85.0%, p < 0.001), reinforcing the link between better awareness and improved adherence to MWM (Table 3). Clean workers at the public hospital outperformed those at the specialized one in awareness scores (68.1% vs. 55.3%, p = 0.004), with adherence

Table 1. Demographic characteristics of the studied groups (N=200)

Variables	Measures	The specialized hospital(n)	The public hospital(n)	p-value
Doctors (n = 80)				
Age (years)	Mean ± SD	30.8 ± 3.2	29.6 ± 4.2	^0.130
	Range	25.0–35.0	25.0–40.0	
Gender	Male (n, %)	24 (60.0%)	25 (62.5%)	#0.818
	Female (n, %)	16 (40.0%)	15 (37.5%)	
Work Experience (years)	Mean ± SD	5.8 ± 3.0	4.7 ± 4.0	^0.179
	Range	1.0–10.0	1.0–15.0	
Nurses (n = 80)				
Age (years)	Mean ± SD	33.2 ± 7.4	31.8 ± 5.5	^0.342
	Range	22.0–51.0	22.0–46.0	
Gender	Male (n, %)	12 (30.0%)	11 (27.5%)	#0.805
	Female (n, %)	28 (70.0%)	29 (72.5%)	
Work Experience (years)	Mean ± SD	10.0 ± 4.4	9.1 ± 2.7	^0.236
	Range	4.0–31.0	2.0–18.0	
Clean Workers (n = 40)				
Age (years)	Mean ± SD	39.5 ± 7.4	37.6 ± 3.0	^0.294
	Range	22.0–50.0	32.0–42.0	
Gender	Male (n, %)	8 (40.0%)	7 (35.0%)	#0.744
	Female (n, %)	12 (60.0%)	13 (65.0%)	
Work Experience (years)	Mean ± SD	5.3 ± 2.5	4.5 ± 1.6	^0.243
	Range	2.0–10.0	2.0–8.0	

^Independent t-test. #Chi square test.

Table 2. Comparison between doctors of both hospitals regarding awareness and practice

Variable	Measure	The specialized hospital (Total = 40)	The public hospital (Total = 40)	p-value
Awareness Score %	Mean ± SD	76.4 ± 10.6	81.9 ± 12.1	0.035*
	Range	50.0 – 93.8	50.0 – 100.0	
Awareness Score % category	**50.0% -**	12 (30.0%)	4 (10.0%)	0.025*
	≥75.0%	28 (70.0%)	36 (90.0%)	
Practice Score %	Mean ± SD	75.2 ± 13.0	87.7 ± 11.5	<0.001*
	Range	50.0 – 91.7	50.0 – 100.0	
Practice Score % category	**50.0% -**	14 (35.0%)	5 (12.5%)	0.018*
	≥75.0%	26 (65.0%)	35 (87.5%)	

*Statistically significant at $p < 0.05$.

^Tests: Independent t-test for mean score comparison; Chi-square test for categorical comparison.

Table 3. Comparison between nurses of both hospitals regarding awareness and practice

Variable	Measure	The specialized hospital (Total = 40)	The public hospital (Total = 40)	p-value
Awareness Score %	Mean ± SD	85.0 ± 9.5	90.3 ± 10.7	0.021*
	Range	56.3 – 100.0	75.0 – 100.0	
Awareness Score % category(n,%)	**50.0% -**	6 (15.0%)	0 (0.0%)	0.026*
	≥75.0%	34 (85.0%)	40 (100.0%)	
Practice Score %	Mean ± SD	85.0 ± 15.5	95.0 ± 6.8	<0.001*
	Range	58.3 – 100.0	66.7 – 100.0	
Practice Score % category(n,%)	**50.0% -**	8 (20.0%)	1 (2.5%)	0.029*
	≥75.0%	32 (80.0%)	39 (97.5%)	

*Statistically significant at $p < 0.05$. ^Independent t-test for mean score comparison. §Fisher's Exact test for categorical comparison.

scores also significantly higher (71.7% vs. 55.0%, p = 0.006), indicating better overall compliance with MWM practices (Table 4). Correlation analysis showed no significant associations between adherence or awareness scores and demographic factors such as age or work experience; however, a strong positive correlation was observed between awareness and practice scores across all job categories (Table 5). Additionally, no significant gender-based differences were found in awareness or adherence scores in any role at either hospital (Table 6).

Homogenous groups had the same symbol "a,b" based on post hoc Bonferroni test.

Pearson correlation test was used for all correlations.

Discussion

This comparative study examined the level of awareness and adherence toward medical waste management among healthcare workers in two hospitals with differing administrative structures. The findings revealed notable differences in both awareness and adherence, with the public hospital significantly outperforming the specialized one across all staff categories. The explanation for this is that the public hospital is a fever hospital, which deals on a daily basis with high risk infectious diseases. Such continuous exposure necessitates strict adherence to infection prevention and control protocols and reinforces awareness through repeated

Table 4. Comparison between clean workers of both hospitals regarding awareness and practice

Variable	Measure	The specialized hospital (Total = 40)	The public hospital (Total = 40)	p-value
Awareness Score %	Mean ± SD	55.3 ± 14.4	68.1 ± 12.3	0.004*
	Range	37.5 – 87.5	50.0 – 93.8	
Awareness Score % category (n.%)	25.0% - (n, %)	6 (30.0%)a	0 (0.0%)b	0.030*
	50.0% - (n, %)	11 (55.0%)a	14 (70.0%)a	
	≥75.0% (n, %)	3 (15.0%)a	6 (30.0%)a	
Practice Score %	Mean ± SD	55.0 ± 17.2	71.7 ± 18.6	0.006*
	Range	33.3 – 91.7	41.7 – 91.7	
Practice Score % category (n,%)	25.0% - (n, %)	7 (35.0%)a	3 (15.0%)a	0.034*
	50.0% - (n, %)	9 (45.0%)a	5 (25.0%)a	
	≥75.0% (n, %)	4 (20.0%)a	12 (60.0%)b	

*Statistically significant at $p < 0.05$. ^Independent t-test for mean score comparison. #Chi-square test, §Fisher's Exact test for categorical comparison. Homogenous groups had the same symbol "a,b" based on post hoc Bonferroni test.

Table 5. Correlations of awareness and practice Scores among Jobs in both Hospitals

Hospital	Variable	Measure	Doctor (Total=40)	Nurse (Total=40)	Clean Worker (Total=20)
The specialized hospital					
Age (years)		r	-0.107	0.254	0.051
		p-value	0.511	0.114	0.831
Work experience (years)		r	-0.067	0.266	-0.353
		p-value	0.682	0.096	0.127
Practice score %		r	0.867	0.931	0.919
		p-value	<0.001*	<0.001*	<0.001*
The public hospital					
Age (years)		r	0.021	-0.196	-0.159
		p-value	0.896	0.227	0.504
Work experience (years)		r	0.007	0.156	0.001
		p-value	0.966	0.336	0.999
Practice score %		r	0.625	0.774	0.887
		p-value	<0.001*	<0.001*	<0.001*
The specialized hospital					
Age (years)		r	-0.041	0.266	-0.066
		p-value	0.803	0.097	0.784
Work experience (years)		R	0.007	0.297	-0.439
		p-value	0.968	0.063	0.053
The public hospital					
Age (years)		R	-0.055	-0.150	-0.127
		p-value	0.735	0.354	0.467
Work experience (years)		R	-0.075	0.307	0.173
		p-value	0.645	0.054	0.467

*Statistically significant at $p < 0.05$. Pearson correlation test was used for all correlations.

Table 6. Comparison according to gender regarding awareness and practice scores % among jobs in both hospitals

Variables	Measures	Doctor (Total=40)	Nurse (Total=40)	Clean worker (Total=20)
The specialized hospital				
Awareness score %	Male	76.6±10.9	84.4±8.2	53.9±12.0
	Female	76.2±10.5	85.3±10.1	56.3±16.2
	p-value	0.911	0.789	0.731
Practice score %	Male	74.7±13.8	84.7±14.1	52.1±13.2
	Female	76.0±12.1	85.1±16.3	56.9±19.7
	p-value	0.745	0.942	0.550
The public hospital				
Awareness score %	Male	82.0±12.9	86.9±11.0	63.4±9.8
	Female	81.7±10.9	91.6±10.5	70.7±13.1
	p-value	0.934	0.222	0.216
Practice score %	Male	85.7±13.1	90.9±10.2	67.9±20.1
	Female	91.1±7.4	96.6±4.2	73.7±18.3
	p-value	0.149	0.101	0.517

Independent t-test.

practical experience. In contrast, the specialized hospital mainly manages general medical cases, including internal medicine, pediatrics, surgery, obstetrics and gynecology, and similar specialties. The absence of statistically significant differences in demographic variables such as age, gender, and work experience between the two hospitals supports the hypothesis that institutional factors, rather than individual characteristics, are the main determinants of awareness and adherence. This is consistent with research by Aboelnour and Abuelela (11), who found that organizational support and continuous training had a more substantial effect on compliance than personal demographics. Similarly, Aslam et al. (12) emphasized that institutional environment specifically the presence of ongoing infectious disease exposure and established infection control culture in public hospitals has a more decisive influence on adherence to medical waste protocols than individual characteristics like age or experience. This underscores the urgent need for specialized hospitals in Egypt to adopt institutional level reforms to close the compliance gap. In Egypt, where healthcare facilities often vary in regulatory oversight and infrastructure quality, this finding suggests that national MW guidelines must be more stringently implemented and audited, particularly in university and private hospitals that may not face the same disease burden as fever hospitals (11,12). In terms of job specific performance, nurses at both facilities showed the highest practice scores, followed by doctors and then clean workers. These findings align with those of Mattoo et al. (13), who noted superior MW practices among nurses, likely due to their direct patient care responsibilities and structured training. Clean workers consistently reported the lowest awareness and adherence, despite showing better scores at the public hospital than at the specialized one. This trend is supported by Anicetus et al. (14), who highlighted waste handlers as the most under informed group in healthcare waste management due to limited training and professional development opportunities. The strong positive correlation between awareness and adherence observed across all job categories reaffirmed the idea that increasing awareness contributes directly to improved compliance with waste management protocols. Conti et al. (15) corroborated this relationship in their meta-analysis, emphasizing the role of multi-component educational interventions in reinforcing both knowledge and behavior. Conversely, the lack of correlation between adherence and demographic variables like age or experience suggests that institutional practices, such as availability of waste segregation tools, regular training, and monitoring, play a more critical role. Khan et al. (16) and Farooq et al. (17) similarly reported that compliance levels are largely influenced by hospital policies and infrastructural support rather than individual experience. Interestingly, gender-based comparisons revealed no significant differences in either awareness or practice scores. This finding supports the assertion made by Alvi et al. (18) and Aboelnour and Abuelela (11) that gender does not significantly influence compliance when institutional training is uniformly applied. However, it contrasts slightly with Vallepalli et al. (19), who observed marginally higher compliance among female nurses, suggesting that gender disparities may be context specific and influenced by educational access or cultural expectations. The overall superior performance of the public hospital can likely be attributed to more consistent implementation of governmental MW regulations and structured oversight mechanisms. These findings highlight the need for university hospitals to adopt more standardized waste management protocols, improve staff training, and enforce compliance through regular audits and feedback mechanisms. This study emphasized the importance of continuous professional development and institutional commitment to safe waste management practices. The integration of awareness campaigns, practical workshops, and systematic monitoring can significantly improve adherence, particularly among non-clinical staff such as clean workers. Additionally, aligning hospital policies with national and WHO guidelines (20) can further enhance the safety and sustainability of medical waste handling systems.

Strengths of the study

One of the key strengths of our study is its comparative cross sectional design, which allowed for a comprehensive evaluation of adherence, and awareness related to medical waste management across two different hospitals. By including a balanced sample size of 200 participants, evenly distributed among doctors, nurses, and clean workers, our study ensured robust statistical power and reliable comparisons between job categories. Additionally, the study's multi-dimensional approach, examining practice and awareness scores alongside demographic factors such as age, gender, and work experience, provided a holistic understanding of the factors influencing medical waste management practices. Another significant strength is the use of standardized assessment tools, ensuring consistency and accuracy in measuring awareness and adherence levels. The inclusion of multiple job categories also allowed for a nuanced analysis, revealing job-specific challenges and strengths. This comprehensive approach contributes to the generalizability of our findings, making them applicable to similar healthcare settings.

Limitations of the study:

Despite its strengths, our study has several limitations. Firstly, its cross sectional design captures data at a single point in time, preventing any causal inferences about the relationship between awareness and practice scores. Longitudinal studies would be necessary to establish causal links and evaluate changes over time. Secondly, the study was conducted in two hospitals located in the same geographical area, which may limit the generalizability of the findings to other regions or healthcare systems. The reliance on self-reported data for awareness and practice scores introduces the potential for social desirability bias, where participants may have over reported positive behaviors or knowledge levels.

Conclusion

Our study concluded that healthcare workers at the public hospital demonstrated significantly higher adherence and awareness scores compared to those at the specialized hospital across all job categories. This indicates more effective implementation of medical waste management practices at the public one. The absence of significant demographic differences suggests that variations in practice and awareness scores were primarily influenced by institutional factors rather than individual characteristics. Positive correlations between awareness and practice scores highlight the importance of knowledge as a key driver of proper medical waste management.

The findings underscore the need for continuous educational interventions and standardized protocols to enhance adherence to medical waste management guidelines. Moreover, the lack of gender-based differences suggests that training programs can be uniformly implemented across all staff members without the need for gender-specific modifications. Our study contributes to the growing body of evidence supporting the importance of awareness and adherence integration in effective medical waste management.

List of Abbreviations:

Abbreviation	Full Term
HCWs	Healthcare Workers
WHO	World Health Organization
MW	Medical Waste
MWM	Medical waste management
SPSS	Statistical Package for the Social Sciences
ANOVA	Analysis of Variance
MOHP	Ministry of Health and Population
IQR	Interquartile Range
SD	Standard Deviation
IRB	Institutional Review Board

Authors' contributions: MHAAG: Conceptualization, study design, data collection, manuscript drafting; NSM: Formal analysis, data interpretation, critical revision of the manuscript. DME: Supervision, methodology guidance, review and editing. MY: Statistical analysis, data visualization, software support. MSG: Project administration, resource acquisition, manuscript review. All authors have read and approved the final version of the manuscript and agree to be accountable for all aspects of the work.

Ethic approval: Ethical approval was obtained from faculty of medicine, Ain Shams University, Research Ethics committee(under the number code FMASU MD 155 /2022 FWA 000017585).

Consent for publication: Informed consent was taken from each participant. The confidentiality and privacy of data were assured.

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Confidentiality of data: The authors declare that they have followed the protocols of their work center on the publication of data from participants.

Declaration on the use of AI: The core intellectual content, critical analysis, data interpretation, and conclusions were entirely developed by the authors without reliance on AI. No AI tool was used to generate research hypotheses, analyze results, or draw scientific conclusions. All authors have reviewed and approved the final manuscript and take full responsibility for the accuracy and integrity of the content.

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